Rakib Hossan

RESEARCHER AND AI ENGINEER

Dhaka, Bangladesh

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Research Interests

Large Language Models (LLMs) and reasoning; claim verification and evidence-grounded fact-checking; abductive reasoning; Natural Language Processing for low-resource languages (Bangla); speech technologies and medical speech recognition; LLM agents and test-time reinforcement learning; reliable, hallucination-resistant AI.

Work Experience

AI RESEARCHER & ENGINEER

Penta Global Ltd. *Dhaka, Bangladesh*

Apr. 2026 - Present

- Work mainly as a researcher on the Penta AI team: Bangla dialectal medical speech recognition, medical prescription understanding, and agentic test-time reinforcement learning (TTRL).
- Design and build LLM-based pipelines for claim extraction, evidence grounding, and dependency-graph validation over scientific documents.
- Apply retrieval-augmented generation, majority-vote consensus across repeated LLM generations, and LLM-judge quality scoring to reduce hallucination and improve output reliability.
- Iteratively debug and validate pipeline stages against real-world documents, tracking failure modes and rejected approaches to keep design decisions evidence-based.

AVNBD *Dhaka, Bangladesh*

Aug. 2026 - Nov. 2026

- Built a multi-stage LLM pipeline to extract, structure, and translate a large-scale human rights dataset from a scanned Bangla source document.
- Cross-referenced extracted records against an independent third-party dataset to validate accuracy and surface discrepancies.

SOFTWARE ENGINEER

Penta Global Ltd. *Dhaka, Bangladesh*

Apr. 2024 - Mar. 2026

- Role prior to transitioning into the AI Engineer position above, at the same company.
- Worked on enterprise-level applications and digital services, including:
  - Bangladesh Election Commission (Web System)
  - ISP Billing Services
  - HRM Management System

OpenCare (Opencarebd Inc.) – Self-employed *Dhaka, Bangladesh (Hybrid)*

Jun. 2025 - Feb. 2026

- Concluded Feb. 2026. Built and shipped the following as part of the team:
- **Features:** Institution management, Keycloak auth, file storage (MinIO), Liquibase, Dockerized deployment.

Contributors: Full list on GitHub
Sadman Sobhan, Aminul Islam, Ismail Hossain, Habib, Tajminur Rahman

Education

Bangladesh University of Business and Technology (BUBT) *Dhaka, Bangladesh*

B.SC. ENGG IN COMPUTER SCIENCE AND ENGINEERING *April 2023 - January 2027*

- Department: Computer Science and Engineering
- CGPA: 4.00 (9th semester running)
- Duration: 4 years

United Marine Academy *Dhaka, Bangladesh*

PRE-SEA TRAINING IN MARINE ENGINEERING *August 2012 - September 2014*

- CGPA: 3.66
- Duration: 2 years
- Group: Marine Engineering

Dhaka College *Dhaka, Bangladesh*

HIGHER SECONDARY SCHOOL CERTIFICATE (HSC) *2012*

- Group: Science
- GPA: 5.00

Dhanmondi Govt. Boys High School *Dhaka, Bangladesh*

SECONDARY SCHOOL CERTIFICATE (SSC) *2010*

- Group: Science
- GPA: 5.00

Ongoing Projects

Thesis + Evidence Claim Pipeline

LLM-BASED NLP RESEARCH *Ongoing*

- **Collaborators:** Shubhashis Roy Dipta, Mir Sazzat Hossain, MD Fahim
- Researching claim verification and abductive reasoning with LLMs, building a system that extracts claims from scientific papers and reasons about the evidence and premises behind them.

Bangla Dialectal Medical ASR Benchmark

MEDICAL SPEECH RECOGNITION RESEARCH *Ongoing*

- **Collaborators:** MD Fahim, Mir Sazzat Hossain, Intesar Tahmid, Galib Mahmud Jim, Md Mahir Jawad
- Building a benchmark to measure clinically significant errors in Bangla dialectal medical speech recognition, using synthetic doctor-patient conversations generated with LLMs and text-to-speech.

Medical Prescription Understanding

MEDICAL DOCUMENT UNDERSTANDING *Ongoing*

- **Collaborators:** MD Fahim, Mir Sazzat Hossain, Intesar Tahmid, Galib Mahmud Jim, Md Mahir Jawad
- Reading and structuring medical prescriptions with AI, so that drug names, dosages, and instructions can be extracted reliably.

Agentic Test-Time Reinforcement Learning

LLM AGENTS RESEARCH *Ongoing*

- **Collaborators:** MD Fahim, Mir Sazzat Hossain, Intesar Tahmid, Galib Mahmud Jim, Md Mahir Jawad
- Applying test-time reinforcement learning (TTRL) to LLM agents, so they can keep improving on new, unlabeled tasks using their own consensus as a reward signal.

Research Publications (FIRST AUTHORED)

AACL-IJCNLP 2025

PROMPTGUARD AT BLP-2025 TASK 1: A FEW-SHOT CLASSIFICATION FRAMEWORK USING MAJORITY VOTING AND KEYWORD SIMILARITY FOR BENGALI HATE SPEECH DETECTION

December 2025

- **Mentor & Co-author:** Shubhashis Roy Dipta
- Focus Areas: hate speech detection, Bengali (Bangla) language, prompt-based classification, LLM-based classification.
- Publication At: ACL Anthology 2025
- **Download PDF:** aclanthology.org/2025

Skills

ML & NLP: CNNs, RNNs/LSTMs, Transformers, text classification, few-shot learning, word embeddings.
LLM Research: LLMs, RAG, prompt engineering, abductive reasoning, LLM-as-judge evaluation.
AI Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn, Pandas, NLTK.
Languages: Python, Java (Spring Boot), TypeScript (NestJS), C/C++, Flutter.
Backend & DevOps: PostgreSQL, Redis, RabbitMQ, Keycloak, Docker, CI/CD, Microservices.

Certifications

Issued by DeepLearning.AI | Coursera

- **Natural Language Processing Specialization** *Apr. 2026*
- **Deep Learning Specialization** *Mar. 2026*
- **Machine Learning Specialization** *Feb. 2026*

Prior Professional Experience (Marine Engineer)

Marine engineering background prior to CS/AI – built discipline in safety-critical, team-coordinated operations.

MT. XIDI, Seacon Star (VLCC, GRT 160000) *At Sea*

JUNIOR ENGINEER *Mar. 2022 - Jan. 2023*

- Maintained main engine, generator (Yanmar), fresh water generator, boiler (MHI/MAC-90 B), and purifier systems.

ATLANTIC 1, Jhampur Group / MT. ARTAVIL, NITC (Super Tankers) *At Sea*

ENGINE CADET *Sep. 2017 - Aug. 2021*

- Maintained main engine (incl. Mitsubishi 7UEC85LS), generator (Yanmar), fresh water generator, S/W treatment plant, purifier, firefighting equipment, and emergency machinery.

References

Available on request.